

Fueling the Future: Establishing a U.S. HALEU Leasing Program to Secure Global SMR Leadership

Jacob Democko

University of California, Berkeley

PUBPOL 247 - Spring 2026

Executive Summary

The world is facing rising energy demand in the face of AI, economic, and population growth that will need to be addressed to prevent climate catastrophe. In response, China, Russia, and the US have started ramping up investments in nuclear energy, seeing it as the path forward to clean, always available, emission free power. These three countries have started to invest in development of new, advanced nuclear reactors known as Small Modular Reactors (SMRs) that are factory-built, smaller scale nuclear reactors that can be deployed in many locations, quickly, and achieve economies of scale rapidly. In addition to their own domestic markets, there is a race for clean energy export dominance, and whichever country can successfully develop their own advanced reactor technology will also be able to export it around the world. This represents a significant crossroads for this nascent technology, where supportive policies may either enable or cripple this new industry.

Currently there are international mechanisms in place through the US Nuclear Regulatory Commission (NRC), the governing body that regulates US nuclear installations, to assist foreign countries in setting up their own nuclear regulating bodies if they do not already have one. In addition, the Export-Import Bank of the US (EXIM) is set up to provide funding for countries that want to purchase US developed SMR technology. These new advanced reactors require a new fuel type called High-Assay Low-Enriched Uranium (HALEU), but there is currently little to no existing domestic supply. Since 2024, the US has banned imports of HALEU from Russia, and started making investment in domestic enrichment capabilities.

For foreign countries that are interested in buying US-developed nuclear energy technology, the fuel and management of fuel waste is the missing piece of the puzzle that this new policy helps to address. The HALEU Asset Leasing Operations (HALO) program is a proposed new legislation that will allow prospective buyers to lease US-made fuel, meaning they will not need to develop their own enrichment capabilities or waste storage. This legislation will position the US as a one stop shop for advanced reactor

exports, help maintain ownership over fuel sources to mitigate proliferation concerns, and help aggregate demand to support domestic HALEU fuel production and clean energy export dominance.

Introduction: Why Nuclear, Why Now?

In the United States, energy demand is set to increase up to 78% by 2050 with rising demand driven by economic growth, population growth, and increased travel that is not offset by increases in energy efficiency improvements. Nuclear power is seeing a resurgence in the United States, with increasing demand for industrial energy driven by data centers (ICF, 2024).

Currently, large scale reactors are the primary supply of nuclear energy in the US, but small modular reactors (SMRs) are emerging as a new category that may expand the feasibility of nuclear energy for a variety of use cases. SMRs seek to provide a reliable, clean source of energy and one large benefit is that they are designed to be built partly or completely in a factory rather than constructed on-site like traditional large reactors. This means that SMRs will be quicker to exploit economies of scale and lead to faster production, decreasing the time required to achieve price and performance parity with existing energy sources (Information Technology and Innovation Foundation, 2025). As part of this expanding nuclear strategy, the US seeks to increase nuclear energy and technology exports to other nations as “clean energy diplomacy” and build global nuclear dominance (Climate Leadership Council, 2024).

Because SMRs are a new technology, there are risks associated with their development and deployment. The risks fall broadly into four categories: technology risks, market risks, regulatory risks, and political risks. Nuclear energy installations in the US and across the world are especially risky due to regulatory challenges, and fuel waste, reprocessing, and storage. SMRs propose a new regulatory difficulty because there are ongoing challenges to how they should be treated under the Atomic Energy Act of 1954, and how the Nuclear Regulatory Commission (NRC) should treat them with licensing requirements (Bipartisan Policy Center, 2024).

Additionally, the High-Assay Low-Enriched Uranium (HALEU) lacks an existing domestic market, and imports from Russia are geopolitically strained and have been banned since 2024. Support from the NRC

may ultimately determine the market success - or failure, of US advanced reactor development. To help mitigate market risks of SMRs, the Export-Import Bank of the US (EXIM) provides financing for SMRs to nations interested in buying US-made SMRs (Export-Import Bank of the United States, 2024). To further support this technology export, this memo proposes that the DOE and NRC should provide support for HALEU fuel leases to prospective US advanced reactor buyers.

Background: Existing Agencies

The Nuclear Regulatory Commission (NRC) was created as an independent agency by Congress in 1974 to regulate the use of radioactive materials for beneficial civilian purposes while protecting people and the environment. The NRC regulates nuclear power plants and nuclear materials through licensing, inspection, and enforcement of its requirements (U.S. Nuclear Regulatory Commission, 2024). The NRC is concerned with radiation protection and reactor safety. It is a primary concern to prevent a major reactor accident that could threaten public health and safety, so requirements are designed to ensure that a massive release of radiation would not occur. The NRC controls nuclear energy regulations, licensing, commissioning, certification, and operational experience. The Nuclear Energy Innovation and Modernization Act (NEIMA) and ADVANCE Act, passed in 2019 and 2024 respectively, required the NRC to update regulations to expedite licensing for new advanced reactor designs, including small modular reactors ([Congress.gov](https://www.congress.gov), 2024).

In April of 2026, the NRC published a final version of 10 CFR Part 53, “Risk-Informed, Technology-Inclusive Regulatory Framework for Advanced Reactors”, commonly referred to as “Part 53”. This new addition to the regulations introduces new safety standards, flexibility for reactor design based on risk analyses, new security requirements, and features that are intended to accelerate deployment (American Nuclear Society, 2025). The NRC claims that new reactor designs could receive approval in 18 months or less, and the licensing cost could be reduced by half or more given the new rules and shorter

review period. Additionally, to fulfill its obligations under the Non-Proliferation Treaty (NPT), the NRC International Regulatory Development Partnership (IRDP) helps foreign countries establish and improve their nuclear safety and security regulatory authorities (U.S. Nuclear Regulatory Commission, 2026).

The Export-Import Bank of the United States (EXIM) was established in 1934 as the official export credit agency of the US, providing financial assistance and supporting exports of US goods and services, especially when private sector lenders are unable or unwilling to provide financing. To support the US nuclear industry, the EXIM will provide financing to nations interested in buying US-made SMRs. To help achieve profitability and economies of scale, it is important that the US leads the push for these reactors and the industry is supported both domestically and internationally. New policies to support financial and licensing support to prospective international reactor buyers would be a boon to the domestic advanced reactor industry.

The US Atomic Energy Act establishes “123 Agreements” which allow the US to cooperate with allied countries in technical exchanges, scientific research, and transfers of nuclear material and equipment. In order to enter into one of these agreements, “that partner must adhere to a strong set of nonproliferation requirements” (U.S. Department of Energy, National Nuclear Security Administration, 2025). The US State Department, with concurrence of the DOE/NNSA and the NRC negotiate and administer these agreements, which will need to be in place for any nation that wants to purchase US SMRs or participate in the new nuclear fuel leasing program, HALEU Asset Leasing Operations (HALO).

Current Approaches

Currently, High-Assay Low-Enriched Uranium (HALEU) fuel is a supply-side bottleneck for SMR development in the USA. Historically, the world relied primarily on Russian HALEU exports, however in 2024 the US banned Russian uranium imports and in 2026 the DOE funded a \$2.7 billion investment in domestic uranium enrichment (U.S. Department of Energy, 2026). Part of the complication with the

market for HALEU fuel is a chicken and egg problem. New advanced reactors will rely on HALEU fuel, but without an existing market incentive there is low financial motive for companies to invest in the costly and energy intensive enrichment facilities that would be needed for robust domestic supply.

Traditionally, uranium used in large water reactors is enriched to 5% U-235, meaning it is 5% U-235 by mass, and the remaining mass is largely U-238, which is the most abundant natural form of uranium. By contrast, HALEU is enriched to 20% U-235. Existing enrichment technology can produce HALEU but plants will need to be updated and more advanced technology used to make it feasible. The other concern with using fuel at higher enrichment levels is the potential for nuclear weapons proliferation.

Weapons-grade uranium is considered to be uranium enriched above 90% U-235. The amount of separative work units required to enrich uranium from 20% to 90% is non-linear, and is substantially lower than enriching natural uranium to 20% U-235 (World Nuclear Association, 2024). For this reason, HALEU needs stricter controls to prevent weaponization.

Currently without a US-based supply of fuel, if a country was interested in buying a US designed SMR, they could work with the NRC to develop regulations and site licensing, but when it comes to the fuel they would be reliant on Russian fuel exports, diminishing the influence and control the US has over the nuclear installation. New investments in uranium enrichment are a good step, but increasing the overall size of the market by introducing fuel leases would help the market achieve equilibrium that makes it more financially feasible to stand up new domestic enrichment capabilities.

The \$2.7 billion in DOE funding for domestic uranium enrichment is a good start to kick off domestic production that has been historically slow, but the primary structural issue is that there is a large gap between the current demand of HALEU fuel and the projected future demand. This makes it a risky proposition for any company to set up enrichment facilities, intended to operate for decades, not to produce small intermittent orders for demonstration or pilot reactors. By selling fuel leases to foreign countries, the US can aggregate larger demand and have the DOE purchase the fuel itself to guarantee and

secure the market. This will bridge the “valley of death” that exists with the high upfront capital costs of new enrichment facilities.

Proposed Policy: HALEU Asset Leasing Operations (HALO)

This proposal is for the Department of Energy (DOE), with the support of the NRC, to stand up capabilities to loan HALEU fuel to buyers of US-made advanced nuclear reactors. The HALEU Asset Leasing Operations (HALO) program would provide HALEU fuel as a lease, under ownership of the DOE/NNSA, to prospective buyer nations. In order to support the budding US advanced reactor industry, providing reactor fuel as a loan will achieve three primary objectives:

1. Position the US as a one stop shop for advanced reactor exports, licensing, and HALEU fuel
2. Maintain ownership over fuel sources to prevent proliferation concerns
3. Support the economic feasibility of both the advanced reactor and domestic HALEU fuel supply by increasing global market share and clean energy export dominance

With this new program, a buyer nation interested in nuclear energy with no existing nuclear infrastructure could come to the US with a proposal, and by working with the NRC International Regulatory Development Program, establish their regulatory agency, the EXIM to fund the purchase of an SMR, and the new HALO program to lease fuel, and they will not need to invest further in nuclear waste management. This comprehensive plan makes it so that the US will be an international leader in nuclear energy exports and clean energy.

The program would work on a 30-year lease term of the fuel, with the US DOE responsible for international shipment, reactor fueling, and refueling and decommission of spent fuel. All of the existing international mechanisms are in place for this program, and now is the right time to kick off the US domestic enrichment program. Leasing the fuel means that the US maintains ownership over the technology, and provides additional security to the purchasing nation because the US maintains a stake in their nuclear facility. The buyer will pay for the shipping which will be handled by DOE expert teams, and when the fuel is returned to the US it will be reprocessed and moved to long term cask storage.

Constituencies, Barriers, and Roadblocks

Nuclear waste storage remains a roadblock in the United States, with no permanent long-term storage facility in operation. The Yucca Mountain project was designated in 1987 as a long-term storage facility in Nevada, but a series of legal and political challenges have delayed its implementation (U.S.

Environmental Protection Agency, 2024). In order to accept HALEU waste that is returned to the US, this will need to be addressed. Nuclear storage is contentious because US states such as Texas see it as hazardous, and Nevada, without any operational nuclear reactors, also does not want to accept the waste (Rabe, B. G., 2026). If a lease program is implemented, the US will need to determine where to store the waste at the end of the term. As part of the HALO proposal, long term storage will need to be developed at the Yucca Mountain site, but because the leases will not expire for 30 years there is sufficient time to develop this site as needed.

The US also faces competition from Russia's state owned nuclear enterprise, Rosatom. Rosatom currently dominates the local nuclear export market, with a large vertically integrated nuclear enterprise including fuel, reactors, and weapons. They hold a large portfolio and are currently constructing 20 nuclear reactor units across 7 countries. This is a risk for the US that will need to prove a competitive advantage, otherwise buyers will find it easier to work with Russia for their advanced reactors and fuel supply dominance (Nuclear Threat Initiative, 2024). The US can rely heavily on its credibility as a world military and economic leader to encourage states to partner and pursue its technology exports.

The final roadblock is the transportation and delivery of the fuel. This will require trained teams and secure transport during fueling and refueling operations, though this is infrequent. The expertise exists within the DOE/NNSA complex, and as part of the financing package through HALO, the buyer nation will be billed for this personnel and delivery, as well as refueling service at the end of 30 years.

Conclusion

With the resurgence of interest in nuclear power and development of advanced reactors, now is the time for the US to implement the HALEU Asset Leasing Operations (HALO) plan to support technology export of advanced reactors. The benefits of the HALO plan are clear, supporting the economic feasibility of domestic HALEU fuel supply, maintaining ownership of fuel sources to prevent proliferation, and positioning the US as a one stop shop for advanced reactor technology export.

Working in conjunction with the existing EXIM and NRC, the HALO plan is ready to implement. The DOE has recently funded investments in domestic HALEU supply and this new plan will help provide foreign demand to accelerate this emerging market. If the US invests now in exporting their advanced reactor technology, they will be in a strong position to compete with Russia in atomic energy exports in the mid- to long-term.

To ignore this opportunity, the US will be ceding technology and international dominance to Russia and China for the foreseeable future, and this should not only be considered an investment opportunity, but an opportunity in international clean energy diplomacy. If the US acts now to implement the HALO plan, they give the advanced reactor market the strongest chance of competing on the world stage and securing the future of energy.

References

American Nuclear Society. (2025, May). NRC unveils Part 53 final rule. *Nuclear Newswire*.

<https://www.ans.org/news/article-7881/nrc-unveils-part-53-final-rule/>

Bipartisan Policy Center. (2024, January). *Licensing and permitting reforms to accelerate nuclear energy deployment*.

https://bipartisanpolicy.org/wp-content/uploads/2024/01/BPC_Nuclear-Permit-License-Reform-Issue-Brief.pdf

Climate Leadership Council. (2024). *The clean energy export opportunity: Nuclear energy*.

<https://clcouncil.org/blog/nuclear-energy-exports/>

Congress.gov. (2024). *S.870 - ADVANCE Act of 2024*.

<https://www.congress.gov/bill/118th-congress/senate-bill/870>

Export-Import Bank of the United States. (2024). *Small modular reactor (SMR) financing*.

<https://www.exim.gov/policies/small-modular-reactor-financing>

ICF. (2024, March 14). The impact of rapid demand growth on the US power grid.

<https://www.icf.com/insights/energy/impact-rapid-demand-growth-us>

Information Technology and Innovation Foundation. (2025, April 14). *Small modular reactors: A realist approach to the future of nuclear power*.

<https://itif.org/publications/2025/04/14/small-modular-reactors-a-realist-approach-to-the-future-of-nuclear-power/>

Nuclear Threat Initiative. (2024). *State Atomic Energy Corporation (Rosatom)*.

<https://www.nti.org/education-center/facilities/state-atomic-energy-cooperation-rosatom/>

Otani, C., et al. (2026). Hazard and risk analysis framework for nuclear power plant–based integrated energy systems. *Nuclear Science and Technology Open Research*, 4(5).

<https://doi.org/10.12688/nuclscitechnolopenres.17727.1>

Rabe, B. G. (2026, March 17). The politics of nuclear waste management and the divergent paths of the United States and Canada. *Taylor & Francis*. <https://doi.org/10.1080/02722011.2025.2557806>

Stanford University. (2026, January). *SETR 2026: Energy technologies*.

https://setr.stanford.edu/sites/default/files/2026-01/SETR2026_04-Energy_web-260109.pdf

U.S. Department of Energy. (2026, April 30). *U.S. Department of Energy awards \$2.7 billion to restore American uranium enrichment*.

<https://www.energy.gov/articles/us-department-energy-awards-27-billion-restore-american-uranium-enrichment>

U.S. Department of Energy, National Nuclear Security Administration. (2025). *123 Agreements for peaceful cooperation*. <https://www.energy.gov/nnsa/123-agreements-peaceful-cooperation>

U.S. Energy Information Administration. (2023). *EIA projections of energy demand to 2050*.

<https://www.eia.gov/todayinenergy/detail.php?id=56040>

U.S. Environmental Protection Agency. (2024). *Summary of the Nuclear Waste Policy Act*.

<https://www.epa.gov/laws-regulations/summary-nuclear-waste-policy-act>

U.S. Nuclear Regulatory Commission. (2024). *About the NRC*. <https://www.nrc.gov/about-nrc>

U.S. Nuclear Regulatory Commission. (2026). *Rulemaking for Part 53: Risk-informed, technology-inclusive regulatory framework for advanced reactors*.

<https://www.nrc.gov/reactors/new-reactors/advanced/modernizing/rulemaking/part-53>

World Nuclear Association. (2024). *High-assay low-enriched uranium (HALEU)*.

<https://world-nuclear.org/information-library/nuclear-fuel-cycle/conversion-enrichment-and-fabrication/high-assay-low-enriched-uranium-haleu>